

Experiment 6: Kafka Integration with C#

Aim:

To create a Chat Application using Apache Kafka as a streaming platform and exchange chat messages between Producer and Consumer applications using C#.

Software Requirements:

- Visual Studio 2022
 - .NET 8 SDK
 - Java JDK 17
 - Apache Kafka
 - Confluent.Kafka NuGet Package
-

Step 1: Install Java

1. Download Java JDK 17.
2. Install Java.
3. Open Command Prompt.
4. Verify Java installation.

`java -version`

Expected Output:

`java version "17.x.x"`

Step 2: Download Apache Kafka

1. Download Apache Kafka Binary Package.
2. Extract the downloaded file.
3. Rename the extracted folder as **Kafka**.
4. Move the folder to

`C:\Kafka`

Step 3: Start ZooKeeper

1. Open Command Prompt.

2. Navigate to Kafka folder.

```
cd C:\Kafka
```

3. Execute the following command.

```
.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
```

4. Wait until ZooKeeper starts successfully.

Do not close this Command Prompt.

Step 4: Start Kafka Server

Open another Command Prompt.

Navigate to Kafka folder.

```
cd C:\Kafka
```

Execute

```
.\bin\windows\kafka-server-start.bat .\config\server.properties
```

Kafka Server starts successfully.

Keep this Command Prompt open.

Step 5: Create Kafka Topic

Open another Command Prompt.

Navigate to Kafka folder.

```
cd C:\Kafka
```

Execute

```
.\bin\windows\kafka-topics.bat --create --topic chat-message --bootstrap-server localhost:9092
```

Verify Topic

```
.\bin\windows\kafka-topics.bat --list --bootstrap-server localhost:9092
```

Output

```
chat-message
```

Step 6: Create Producer Project

1. Open Visual Studio.

2. Click Create a New Project.
3. Select Console App (.NET).
4. Click Next.
5. Project Name

KafkaProducer

6. Click Create.

Step 7: Install NuGet Package

Right Click Project

→ Manage NuGet Packages

Search

Confluent.Kafka

Click

Install

Step 8: Add Producer Code

Open Program.cs.

Replace the existing code with Producer code.

Save the project.

Step 9: Run Producer

Press

Ctrl + F5

Output

Kafka Chat Producer Started

Type message and press Enter.

You:

Step 10: Create Consumer Project

Create another Console App.

Project Name

KafkaConsumer

Step 11: Install NuGet Package

Manage NuGet Packages.

Install

Confluent.Kafka

Step 12: Add Consumer Code

Replace Program.cs with Consumer code.

Save the project.

Step 13: Run Consumer

Press

Ctrl + F5

Output

Kafka Chat Consumer Started

Waiting for messages...

Step 14: Test Chat Application

Type a message in Producer.

Example

Hello

Producer Output

Message sent to Kafka

Consumer Output

Received : Hello

Type another message

Hi

Consumer Output

Received : Hi

Step 15: Test Multiple Messages

Producer

Kafka Demo

Consumer

Received : Kafka Demo

Step 16: Stop Applications

Close Producer.

Close Consumer.

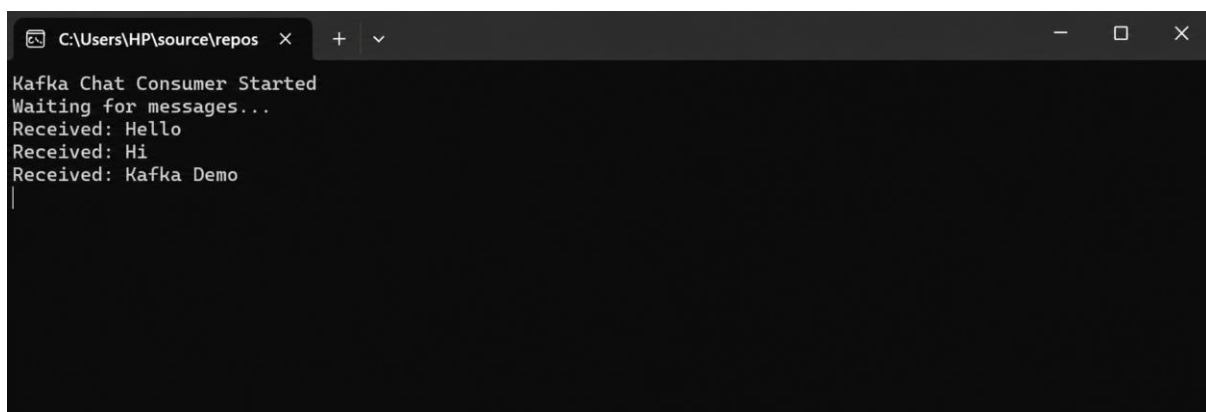
Close Kafka Server.

Close ZooKeeper.

Result

Apache Kafka was successfully installed and configured. ZooKeeper and Kafka Server were started successfully. A Kafka topic named **chat-message** was created. The C# Producer application successfully published chat messages, and the C# Consumer application received the messages in real time using the **Confluent.Kafka** library. Thus, Kafka integration with C# was implemented successfully.

Outputs:



```
C:\Users\HP\source\repos x + v - □ x
Kafka Chat Consumer Started
Waiting for messages...
Received: Hello
Received: Hi
Received: Kafka Demo
```

```
C:\Users\HP\source\repos x + v
Kafka Chat Producer Started
Type message and press Enter. Type 'exit' to stop.
You: hi hello how are u
Message sent to Kafka
You: |
```

Containers

Images

Volumes

Kubernetes

Builds

Models

MCP Toolkit BETA

Docker Hub

Docker Scout

Extensions

Images / apache/kafka:3.9.2

apache/kafka:3.9.2 IN USE

05b4616e0702

CREATED
5 months ago

SIZE
626.65 MB

Recommended fixes ▾

Run ▾

Layers (24)

Vulnerabilities

Packages

0

ADD alpine-minirootfs-3.23.3-x86_64...

9.11 MB

1

CMD ["/bin/sh"]

0 B

2

ENV JAVA_HOME=/opt/java/openjdk

0 B

3

ENV PATH=/opt/java/openjdk/bin:/u...

0 B

4

ENV LANG=en_US.UTF-8 LANGUAGE...

0 B

5

RUN /bin/sh -c set -eux; apk add --no...

37.53 MB

6

ENV JAVA_VERSION=jdk-21.0.10+7

0 B

7

RUN /bin/sh -c set -eux; ARCH="\$(ap...

164.52 MB

8

RUN /bin/sh -c set -eux; echo "Verifi...

12.29 KB

Analyzed by **docker scout**

This image has not been analyzed

You can use Docker Scout to analyze local images and list its vulnerabilities.

Start analysis

[Enable background indexing in Settings](#) so your results are always ready.

Engine running

RAM 2.02 GB CPU 2.34% Disk: 4.02 GB used (limit 1006.85 GB)

Update available